



Infection Prevention and Management Manual

Scabies, Lice, Fleas and Bedbugs Guideline

Scope

Site	Service/Department/Unit	Disciplines
SCGOPHCG	Organisation Wide	All Staff

1. Scabies (*Sarcoptes scabiei*)

Scabies is an infestation of the skin caused by a small mite - *Sarcoptes scabiei*, which is just visible to the naked eye (0.2-0.4mm long). The mite that causes scabies is specific to humans and is able to survive only 2 to 3 days off the human body. The female mite burrows into the outer layers of the skin where it lives and lays its eggs.

Transmission

Transmission is usually by direct and prolonged (at least 20 minutes) skin-to-skin contact. Indirect spread from bedding, clothing and fomites is less common. The risk of transmission from both skin-to-skin contact and fomites is higher with people who have crusted scabies (see below).

Incubation period

The incubation period for people without previous exposure to scabies is 2 - 6 weeks. Shorter incubation periods of 1 to 4 days may be seen with reinfestation.

Signs and symptoms

An allergic reaction to the mite causes the signs and symptoms of scabies. Scabies mites cause intense itching, especially at night or after a hot bath/shower. The presence of mite infestation is evident as papules, vesicles or tiny linear burrows. Thread-like tunnels may be visible as grey lines in the skin. Burrows may occur anywhere but are mainly in skin flexures on the hands and arms, and particularly in the finger webs. Genitalia may also be affected. In adults, the face and neck are rarely affected. The rash usually appears in the groin, elbows, waist, buttocks, shoulder blades, hands and wrists and does not necessarily correspond to the position of the mites.

Crusted scabies

Crusted scabies is a hyper infestation with a much higher burden of mites and is consequently more transmissible. Crusted scabies usually occurs in immunocompromised persons and presents with skin plaques, scale and hyperkeratosis, especially of the hands, feet and scalp. Itch may not be present.

Diagnosis

The diagnosis of scabies is usually clinical. Diagnosis of scabies can be confirmed by obtaining a skin scraping and should be sought in all suspected cases in hospital. However, a negative result on skin scrapings does not rule out infestation because the infected person may have few mites, and these can be easily missed.

Management of patients with suspected or confirmed scabies

- A single room and Standard and Contact Precautions (orange sign- long sleeved gown) are required for cases of suspected and confirmed scabies.
- The Infection Prevention and Management (IPM) Unit should be informed of all suspected or confirmed scabies cases.
- Refer to Australian Therapeutic Guidelines for treatment guidance (available on the SCGOPHCG Intranet Library).
- For non-crusted scabies, Contact Precautions are to continue for 24 hours after initiation of treatment.
- For crusted scabies, Contact Precautions are to continue until all treatment is complete and at least 10 days have passed since treatment initiation.
- Bed linen and clothing should be changed before and after the application of the topical medication. Place all linen in a plastic bag and then in the usual laundry bag.
- Note scabies treatment is generally repeated after 7 days.
- Household contacts and partners may also require treatment to prevent reinfestation, even in the absence of symptoms. In crusted scabies, the mite population on the patient is high and empiric treatment is recommended for household and close contacts.
- Even after an effective course of treatment, the itch associated with scabies can take weeks to resolve.

Management of staff contacts

- Staff members who have had direct or indirect contact with a patient with scabies while not wearing appropriate PPE should be identified and their details provided to Health Safety and Wellbeing (HSW) and IPM Unit.
- Staff who have had casual contact (i.e. less than 20 minutes) with a patient with non-crusted scabies while not wearing appropriate PPE may continue to work.
- Staff who have had prolonged, direct skin-to-skin contact (i.e. more than 20 minutes) with a patient with non-crusted scabies **OR** brief direct or indirect contact (i.e. with clothing/bedding) with a patient with crusted scabies while not wearing appropriate PPE are at higher risk; they should be treated along with their household contacts (if relevant), and remain off work until 24 hours after initiation of treatment.
- Staff contacts, whether treated or not, should self-monitor for symptoms and seek medical advice if symptoms develop, as well as informing HSW and the IPM Unit.
- The IPM Unit must be informed of any suspected hospital transmission of scabies.
- A suspected or confirmed outbreak of scabies on a hospital ward may require additional measures, e.g. additional cleaning, contact precautions and/or treatment of asymptomatic ward contacts, treatment of staff household contacts, ward closure. This should be coordinated by the IPM Unit or on-call Clinical Microbiologist.

2. Head Lice (*Pediculus humanus var. capitis*)

Head lice are parasitic insects, usually confined to the scalp but may occur in the axillae, beard, eyelashes and eyebrows. Infestation is known as pediculosis capitis and can cause significant itching of the scalp and the nape of the neck, due to an allergic reaction to the louse bites. Adult lice are 2-3mm in size, and firmly attach eggs to the base of the hair. These white eggs are visible as 'nits'. Head lice are transmitted by head-to-head contact, sharing combs or headgear. Lice cannot jump or fly and are not spread by pets.

Diagnosis and management of head lice

- Patients with suspected or confirmed head lice are managed with standard precautions.
- Refer to the Australian Therapeutic Guidelines for advice on diagnosis and management of head lice.
- Usually, diagnosis is confirmed by visualisation of crawling lice after combing with a fine-toothed comb.
- If crawling lice are not seen, finding nits attached to the base of hair shafts suggests but does not confirm infestation.
- Bed linen and clothing should be washed on a hot cycle (60°C)
- Household members and close contacts should be examined and treated if head lice infestation is evident.

3. Body Lice (*Pediculus humanus var. corporis*)

Body lice are slightly larger than head lice (2-4mm) and are found mainly in clothing, travelling regularly to the body to feed. Itching is the main symptom. Body lice are able to survive for up to a week off the host without feeding. Transmission is by direct contact with an infested person or clothing, or personal belongings used by them.

Diagnosis and management of body lice

- Patients with suspected or confirmed body lice are managed with standard precautions.
- Diagnosis is confirmed by finding lice or their eggs in seams of clothing or, less commonly, lice crawling or feeding on the skin.
- Treatment is as for head lice (see Australian Therapeutic Guidelines).
- Bedclothes and clothing should be washed on a hot cycle (60°C) or discarded.

4. Pubic Lice (*Phthirus pubis*)

Pubic lice are short and crab-like. They are usually found in pubic hair but may also be found in axillae, beard and eyebrows. Itching in the pubic and groin area is the most common symptom of pubic lice infestation. Pubic lice can survive for only a short time away from the host. They are transmitted by direct, usually sexual, contact. Transmission via fomites such as clothing and bed linen may occur but is thought to be uncommon.

Diagnosis and management of head lice

- Patients with suspected or confirmed pubic lice are managed with standard precautions.
- Diagnosis is confirmed by finding crab-like lice or nits in the hair in the pubic regions.
- Treatment is as for head lice (see Australian Therapeutic Guidelines).
- Shaving pubic hair is beneficial.
- Bedclothes and clothing should be washed on a hot cycle (60°C).
- Household members and sexual partners may also require treatment.

5. Fleas

Like lice, fleas are small flightless insects that feed on the blood of animals. The most common fleas in Australia are the cat flea and the dog flea. Fleas usually live on non-human animals, but may survive for months in the environment (e.g. on clothing, bedding, carpets). Fleas do not live on humans but may bite them opportunistically. Bites are usually extremely itchy and may become secondarily infected. A number of infectious diseases may be transmitted by fleas (e.g. murine typhus, bubonic plague) but most are

extremely uncommon. In the absence of clinical evidence of an infection, no additional investigations are required and prophylaxis is not indicated.

Management of suspected or confirmed flea infestation

- Patients are managed with standard precautions.
- Treatment of uncomplicated bites is symptomatic (e.g. anti-pruritic).
- The patient should shower or be washed.
- Bedclothes and clothing should be washed on a hot cycle (60°C).
- Pest control may be required to treat the household environment. Contact Facilities Management for environmental control in hospital.
- Household pets may need to be treated by a veterinarian.

6. Bed Bugs

Bed bugs are small, parasitic insects that live in the household environment (e.g. carpets, mattresses) and feed on the blood of people and animals while they sleep. Bed bugs may be transported from place to place as people travel, in the seams and folds of luggage, bags, folded clothes and anywhere else they can hide. In hospital, bed bugs can spread quickly from room to room as they can crawl long distances. The management of bed bugs relies on eradicating them from the environment. Bed bugs are not known to spread disease.

Clinical evidence of bed bug infestation

- Bite marks on face, neck, arms, or any other body parts while sleeping. Bite marks may take as long as 9 to 14 days to develop in some people.
- Skin reactions include redness, swelling and lumps approximately 2cm in size, accompanied by itching and burning sensations

Management of suspected or confirmed bed bug infestation

- Place patient in a single room on standard precautions.
- Inform the IPM Unit of any suspected or confirmed case of bed bugs.
- Facilities Management should also be informed regarding treating mattresses, furniture and the environment.
- The patient should shower or be washed.
- Anti-pruritic medication may be required.
- Bed linen should be placed inside a plastic bag and sent for laundering on a hot cycle (60°C).
- Put patient's belongings in a sealed plastic bag and ask the relatives to take the patients old clothes home to be laundered in hot water
- Do not add or remove any furniture from the room.
- Unless the surfaces are non-intact, there is no need to dispose of mattress or pillow in a hospital setting as these items are covered in plastic and can be easily cleaned.
- Close the room until Facilities Management determines it is safe to use.
- Monitor other patients on the ward for signs and symptoms of bed bug bites.

Related Documents

[SCGOPHCG IPM Manual](#): Standard and Transmission based precautions Policy.

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